25900672 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course(VAC) - MG3VACTXLA04 - TEXTILE PROCESSING FOR SUSTAINABILITY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Recognize the method that reduces transportation-related emissions. [K] / [CO1]

   a). Closed-loop systems b). Water conservation

   c). Sustainable finishing d). Local production

2. Which biodegradable textile is recognized as a fast-growing renewable resource? [K] / [CO1]

   a). Bamboo b). Wool

   c). Rayon d). Polyester

3. Which of the following is NOT a benefit of recycling and upcycling in textiles? [K] / [CO1]

   a). Reduced waste b). Conserved resources

   c). Increased landfill use d). Economic opportunities

4. The method of designing clothes so that fabric waste is minimized is called: [U] / [CO1]

   a). Fast fashion b). Upcycling

c). Zero-waste pattern cutting d). Digital printing

5. Recognize the sustainability dimension that increases brand reputation and loyalty. [K] / [CO1]

a). Environmental benefits b). Social responsibility

c). Economic advantages d). Production efficiency

6. Printed textiles used for branding and marketing come under. [U] / [CO3]

a). Home Décor b). Industrial Textiles

c). Advertising and Promotion d). Interior Design

7. Which dyeing method is ideal for large-scale continuous production? [K] / [CO2]

a). Continuous dyeing b). Pad dyeing

c). Batch dyeing d). Garment dyeing

8. Which of the following in not a benefit of natural dyeing? [K] / [CO2]

a). Eco-friendly b). Biodegradable c). Non-toxic d). Low colour fastness

9. Dyeing plays a major role in textile because: [E] / [CO3]

a). It adds aesthaetic value b). It increased market demand c). iIt allows variety and fashion trends d). All of the above

10. Natural dyes are considered sustainable because they are: [U] / [CO3]

a). Petroleum based b). Biodegradable and renewable c). Chemically toxic d). Non-renewable

[1x10 = 10]

Part B

Fill In The Blanks

Answer all questions

Each question carries **1** marks

11. Printing detailed, complex designs with high precision is a key advantage of _____ printing. [U] / [CO1]

12. Sustainability in Textiles aims to minimize _____, social, and economic impacts. [U] / [CO1]

13. Promoting _____, human rights, and community well-being is the focus of social sustainability. [U] / [CO1]

14. Sustainable textile production can create new job opportunities in areas like _____ and upcycling. [U] / [CO1]

15. Waste reduction in textile production helps lower _____ pollution. [U] / [CO1]

16. _____ cotton is grown without toxic pesticides or synthetic fertilizers. [U] / [CO1]

17. The natural origin of dyes makes them _____ and _____. [U] / [CO2]

18. Dyeing fibers before they are spun into yarn is called _____ dyeing. [U] / [CO2]

19. In block printing, _____ blocks are curved with designs. [K] / [CO3]

20. Shade _____ is a common problem in natural dyeing. [U] / [CO3]

[1x10 = 10]

Part C

One Word Questions

Answer all questions

Each question carries **1** marks

21. What property of nano technology textiles protects skin from sunlight? [K] / [CO1]

22. Which type of cotton is most commonly used in sustainable fashion? [K] / [CO3]

23. What do we call making bags or accessories from old clothes? [U] / [CO1]

24. What certification evaluates recyclability and non-toxicity of textile products? [U] / [CO1]

25. What practice involves transforming waste materials into better-quality products? [K] / [CO1]

26. Identify the practice that prevents child and forced labour. [U] / [CO1]

27. What is the characterestics of natural dyes? [K] / [CO2]

28. Which factor controls how deeply the dye enters the fabric? [K] / [CO2]

29. What aspect of eco-friendly fashion focuses on honesty in supply chains? [K] / [CO3]

30. State the printing method uses hand-carved blocks. [K] / [CO3]

[1x10 = 10]

Part D

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

31. Recall how dyeing in textiles leads to water pollution. [K] / [CO1]

32. What are the major challenges in adopting waterless dyeing technologies? [U] / [CO1]

33. Define sustainable textile finishing. [K] / [CO1]

34. What happens during the dyeing stage? [K] / [CO2]

35. How does dyeing contribute to the functionality of textiles? [U] / [CO2]

36. What is the role of responsible sourcing in eco-friendly fashion? [K] / [CO3]

[2x5 = 10]

Part E

Long Answer Questions

Answer any **2** questions

Each question carries **5** marks

37. Discuss the various purposes of textile printing and explain how it serves both decorative and functional roles. [U] / [CO3]

38. Explain how each of the 4 R's—Reduce, Reuse, Recycle, and Recover —contribute to sustainability. [U] / [CO1]

39. How do nanotech and enzyme-based processing aid sustainable textiles? [U] / [CO1]

40. Discuss the future of eco-friendly fashion, highlighting key trends, impacts, and challenges that shape its role in the fashion industry. [U] / [CO3]

[5x2 = 10]